

JAY CHAK

Quality Assurance Engineer (Amazon) | AI & Full-Stack Product Builder | Founder, LeadShala

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SUMMARY

Quality Assurance Engineer with **4.5+ years at Amazon** (Appstore & Fire TV) who also designs, builds and ships AI products end to end. At Amazon, built **Hawkeye**, an agentic-AI IP-infringement analyzer that cut a 30–60 minute specialist review to ~3 minutes, and **Pulse**, a real-time audit workspace that replaced the team's weekly Excel workbooks; logged **350+ CX disparities** and **60+ PO defects** across five competitive-benchmarking cycles. Outside Amazon, built **six products with AI agents in the loop**: Elyxis (n-of-1 experimentation platform, ~1,800 automated tests), I'm Okay (family safety app in 40 languages), GlassBox (live AI browser agent), Patchbay, Postmark, and Cue (real-time AI call copilot, in build with a team). Open-source contributor with **3 PRs merged upstream** (career-ops, Signadot CLI) and 4 in review. Plus 6 years of founder-led growth marketing (Google & Meta Ads).

EXPERIENCE

Quality Assurance Engineer (QAE) — Amazon, Appstore & Fire TV

Feb 2022 – Present

Bengaluru, India · App quality, policy compliance, competitive benchmarking, platform integrity, QA automation

- Designed and built **Hawkeye**, an **agentic-AI intellectual-property infringement analyzer** for app-store submissions: given one app identifier it pulls the complete listing record (developer details, title, description, artwork, package name, prior reviewer notes) from an internal catalog service behind SSO, then investigates whether the submitting developer actually owns or is licensed to distribute the app — cross-store comparison, reverse-image provenance on every icon and promotional asset, developer-identity and website triangulation, a keyword-and-title policy audit, and live-only **trademark-register lookups (USPTO and the WIPO Global Brand Database)** matched to the app's own marketplaces — and returns an **evidence-backed verdict (Likely Infringement / Rights Established / Inconclusive)** with click-to-verify proof for every finding and a generated PDF, cutting a review that took a specialist **30–60 minutes down to ~2.5–4 minutes**.
- Architected Hawkeye's **three analysis engines from one shared policy playbook**: a **Fast** mode that runs the checks as a parallel evidence graph with four concurrent LLM judges and a first-strike early exit, a **Deep Search** mode that is an autonomous browser agent (plan-act-observe loop, up to 320 steps) which narrates every step and can be paused, steered or run in approve-each-step mode, and a **Bulk** mode that processes up to 500 apps through a worker pool with automatic Fast-to-Deep escalation, per-item retries and resume-after-restart. Built full-stack: **Python / FastAPI / asyncio** backend driving **Playwright** (headless Chromium), a **React + TypeScript** operator console with a live WebSocket run feed, and **Anthropic Claude (Opus + Sonnet) on Amazon Bedrock** orchestrated behind a process-global adaptive throttle gate, prompt caching and truncation-aware retries — with deterministic verdict-integrity guards enforced in code above model output, a **100+-check behavioral test suite** and a known-truth regression harness.
- Built **Pulse**, a real-time web tool that replaced the team's two regional weekly Excel workbooks for Policy / Functional / Feature app audits: an admin uploads each week's workbooks into region-tagged process sheets and grants **per-sheet, per-tester access**; testers work a **spreadsheet-grade editable grid** with inline picklist editors, per-block work assignment, Excel-style cell addressing and **full cell-level change history** (who changed what, when, old and new values); weeks can be locked, and everything **syncs live across concurrent editors over WebSockets** with an admin dashboard, filters, cross-week history lookup and styled Excel export. Engineered the **Python / FastAPI + SQLAlchemy** backend (SQLite in dev, PostgreSQL in production) with JWT auth and role-based access, a **React + TypeScript + Tailwind** frontend, an importer that ingested **six months of history (~10,000 audit rows)** from the legacy workbooks, **120+ automated tests** and a **Docker-to-AWS (ECR, App Runner, RDS)** deployment kit.
- Benchmarked **75+ top streaming apps** on Fire TV against Apple TV, Google TV and Roku across **five quarterly parity cycles**, logging **350+ customer-experience disparities (60+ PO)** in JIRA with the platform team — including a back-button UX study of 20 PO apps that exposed exits taking up to 12 clicks.
- Audited **200+ app policy decisions** for the team's Center-of-Excellence audit function, surfacing **50+ tester mismatches** and coaching peers on recurring error patterns.
- Owned daily quality monitoring for **300+ live news channels** on Fire TV's news platform — playback failures, ingestion SLAs, stale-content thresholds — and cut false-positive tickets by tuning trailing configurations while keeping every escalation within SLA.
- Devised an **APK pre-testing mechanism and SOP** validating Google Play apps' Fire TV compatibility ahead of onboarding — accelerating catalog-gap closure inside a selection program credited with **\$2M+ revenue impact**.
- Uncovered a **paid-app scam pattern** among YouTube channel-creator apps, triggering verification of every paid CC app; contributed to platform-integrity enforcement that disabled **4,000+ fraudulent vendor accounts** and suppressed **16,000+ non-compliant apps**, including 1,263 illegal IPTV streams.
- Contributed to the team's QA automation drive — **Maestro UI automation** across the top-40 catalog apps, an automated results pipeline covering **500+ apps (~85% manual-effort reduction)**, and review-processing automation cutting per-app handling from **10–15 minutes to 6.3 seconds** — and supported SDK compatibility analysis across the top-200 app list.
- Resolved **100+ third-party developer cases** end-to-end in Amazon's partner CRM; built **7+ operational SOPs** (including their HTML backends), the team newsletter, and training visuals that reduced decision mismatches across the team.
- Sustained **100% quality scores**, surpassing team averages for consecutive months; earned multiple **Spot Awards and Jedi Ranger Awards**.

Founder & CEO — LeadShala Digital Media

Aug 2020 – Present

Indore, India · MSME-registered performance-marketing company; parent company of Elyxis and I'm Okay

- Founded and ran a performance-marketing business end-to-end for **6+ years** — niche selection, cold outreach, offer design, custom landing pages, and **Google & Meta ad campaigns** — for local service businesses across India, the US, UK and Australia.
- Sold with a 30-day money-back guarantee and ran every function solo: prospecting, copywriting, creative, media buying and reporting — earning client recommendations for communication and problem-solving.
- Built a **28,000+ follower** LinkedIn audience; Meta Blueprint and Google Ads certified.

PROJECTS

Cue — Real-Time AI Call Copilot (meetings, sales calls, interview prep)

Sep 2026 – Present

In build, team-led · Private repo: github.com/jchak7/cue · TypeScript, WebSockets, Postgres + pgvector, Redis

- Founding and architecting a **consent-first real-time copilot**: audio in, streaming speech-to-text with provider failover, question detection, context over the user's résumé, job description and documents, then answers streamed into a desktop overlay, web app and browser fallback on macOS and Windows, targeting **<2.5 s time-to-first-token**. Designed an incremental pipeline that starts intent detection and context prefetch on partial transcripts rather than waiting for the final question. The product boundary is written into the spec: a normal overlay, no proctoring or capture-detection bypass, no "undetectable" claims.
- Authored the **Build Bible** (15-phase roadmap, recorded architecture decisions, competitive teardown of the category leader and five open-source reference repos), a Prisma data model and a **squad-by-squad ticket backlog** so a multi-person team and AI agents can build in parallel under one-ticket-one-branch-one-PR rules with CI-gated merges.
- Shipped the **Wave-0 foundation in two days (23 commits)**: Turborepo + pnpm TypeScript monorepo with CI gates, branch protection and CODEOWNERS; Zod env registry and secrets policy; an **authenticated WebSocket realtime gateway** (heartbeat, session state store, session-channel + launch-token contract); a **model gateway** with provider adapters, vendor-neutral model tiers, a question router and per-user budget metering; palette-gated design tokens; Postgres 16 + pgvector / Redis 7 dev stack; privacy defaults.

GlassBox — Transparent, Steerable AI Browser Agent

Aug 2026

Live · github.com/jchak7/glassbox · Python, FastAPI, Playwright, Claude API, Railway

- Built and deployed an **autonomous browser agent** that takes a plain-English goal and executes it in a real browser — an LLM plans and re-plans each step against live page state rather than replaying a hardcoded script. **Concept to public deployment in a single working day, solo.**
- **Transparency and human control were the design thesis**: reasoning, the chosen action and a live page screenshot stream to the UI over **WebSocket** as each step occurs, and the operator can stop, approve a step, or take over mid-run. A timeline rail marks every step green (landed), red (failed) or **blue (human intervened)**, so a non-technical user can follow and steer the run.
- Stack: **Playwright** driving headless Chromium server-side, **Python + FastAPI** running the plan/act loop, WebSocket step-streaming, Claude API as the reasoning layer, deployed on **Railway** with a backend version marker so deploys are confirmed rather than inferred from a cacheable surface.
- Quality: three working task presets (invoice extraction **verified 14/14 rows across two end-to-end runs**), **no silent failures** — every stall is diagnosed and reported in plain language; a **runnable adversarial suite led by prompt-injection tests** (the failure mode specific to browser agents) ships in the repo, alongside a published LIMITATIONS document and an engineering journal. Live at glassbox-production-d5f7.up.railway.app.

Patchbay — Visual Pipeline Builder (node-graph editor for LLM workflows)

Aug 2026

Live · github.com/jchak7/patchbay · React, ReactFlow, Zustand, FastAPI

- Drag-and-drop node-graph editor for LLM workflows, built on a **data-driven node abstraction** — each node type is one declarative object rendered by a shared factory (adding a type is ~8 lines, not a component). **React / ReactFlow / Zustand; Python + FastAPI** backend validates the graph is a **DAG (Kahn's algorithm)** before it runs; dynamic `{{variable}}`-to-input-port binding; token-based dark design, no UI library.

Postmark — Evidence Layer for a Job Search

Aug 2026

v0.5 · Local-first, zero runtime dependencies · Node.js

- Turns employer confirmation emails into receipts carrying the employer's own timestamp — the only externally attested date in the pipeline — with an append-only status ledger and BM25 recall over past answers. 192 automated tests, zero runtime dependencies, binds 127.0.0.1 only.

Elyxis — A/B Testing for Humans (n-of-1 experimentation platform)

Feb 2026 – Present

Live · elyxis.app · React Native, Next.js, Supabase, Python, Claude API, Paddle

- Building an **n-of-1 self-experimentation platform**: AI designs a rigorous protocol for one change at a time (counterbalanced on/off schedules, duration brackets from 10 to 400 days), the user logs a 5-second daily rating, wearables sync as covariates, and a **Bayesian engine returns an honest verdict** — including "no clear effect" when that is the truth.
- Built the entire stack **solo with Claude Code** as AI pair-programmer: React Native + Expo app, Next.js marketing site + operator dashboard, Supabase (EU) Postgres with row-level security + Deno edge functions, stateless Python FastAPI Bayesian statistics service (Railway), Claude-API protocol designer behind safety & banned-claims guards, and Paddle billing.
- Shipped **~1,800 automated tests** (incl. 13 property-based invariants run at 100+ random iterations), **six OWASP-based red-team passes and six ecosystem audits** — the live Paddle payment path closed its final audit with zero criticals; GDPR-grade export/deletion with de-identified, zero-retention statistics. Computed the platform's **first live Bayesian verdict on production data** in July 2026 (99% likely helped, +3.2 points over 12 valid days). Applied to **Y Combinator (Fall 2026)**.

- One-month build telemetry (YC Paxel): **388,577 lines across 604 commits**, 147 sessions, 184 hours — profiled as an “architect-controller” who converts AI output into bounded, verified engineering work.
- **Live demo:** Android build at github.com/jchak7/elyxis-download · demo login: yc-demo@elyxis.app / Test@7024# · 16-phase roadmap at elyxis.app/vision.

I'm Okay — Family Safety Check-ins

Jan 2026 – Present

Final build, Android & iOS · imokay.in · Node/Express, Postgres, Redis, Next.js, Kotlin

- Designed and built a safety check-in app for families in **~6 months alongside a full-time job**, in **40 languages**: users confirm they're okay on a schedule; missed check-ins alert trusted contacts via a plan-aware escalation pipeline (email free, WhatsApp paid). Shipped: caregiver portal, meal/medicine/water tracking, weekly PDF reports over WhatsApp, AI care assistant, Emergency SOS, five paid tiers (₹0–₹299).
- Engineering: Node + Express API on Postgres holding all authority (client never trusted), Next.js PWA + Capacitor store wrappers **plus a fully native Kotlin/Jetpack Compose Android app** (Android throttles PWA background work — the alarm must fire anyway); Redis locking, 18+ background workers, and **exactly-once alert delivery** via a registry-first outbox; server time only. Deliberately calm UX — no countdowns, no tracking, no panic.

OPEN SOURCE

Open-Source Contributions — 3 merged upstream, 4 in review

2026

github.com/jchak7 · JavaScript, Go, TypeScript, docs

- **career-ops** (career-ops-hq/career-ops · 72k stars) — **2 PRs merged**. #2809: +580 lines (parser plus 402 lines of tests) reviewed by two maintainers and CodeRabbit; asserts a fixed-point property, parse(format(x)) stable after one pass, with an opt-in strict mode, byte-identical default path across eight input variants and every fix pinned by mutation testing. #3230: the follow-up the maintainer invited in the merge message — a --read --strict CLI seam wired into apply mode, with tests pinning the contract.
- **Signadot CLI** (signadot/cli, Go) — **merged**. #364: raw carriage returns in --attach text output let CRLF log lines overwrite the terminal; trimmed \r\n and forced quoting of embedded \r to match JSON mode, with 60 lines of new tests.
- **Moss** (usemoss/moss · YC F25) — in review. #513: +5,481 lines across 30 files — moss-obsidian, a local-first semantic-search plugin for Obsidian: hybrid semantic + BM25 retrieval, heading-aware chunking with breadcrumbs, CommonMark-correct fence handling, native runtime isolated in a worker child process.
- **WarpBuild** — in review. snapshot-save #89: the snapshot lookup trusted the first alias the API returned, so a query for my-image could update my-image-arm64; now selects by exact alias match. action-debugger #24: documented watching GUI/e2e sessions live (Xvfb + x11vnc + noVNC) since tmate cannot forward X11.
- **SigNoz** (docs) — in review. #4242: the ECS-on-EC2 daemon guide tagged every app's telemetry with the collector's own task; moved ECS detection to the app SDK and fixed two adjacent config errors. Also filed metorial/metorial #134, a verified rewrite of an out-of-date CONTRIBUTING.md.

SKILLS

AI-Native Development: Claude Code (daily driver) · Antigravity (Google agentic IDE — built I'm Okay in it) · Kiro (spec-driven development: 37 requirements, 13 correctness properties) · Claude Design (20 signed canvases) · Claude API & Gemini in-product integration · LLM guardrails, banned-claims & safety screening · AI-assisted test design · browser agents (Playwright + LLM plan/act loop) · agent orchestration & step streaming · human-in-the-loop agent UX · multi-LLM judges & model gateways · real-time STT-to-LLM pipelines · prompt-injection testing

AI Method & Machinery: Authored 4 AI skills (build conductor · self-evolving red-team engine · 5-phase ecosystem audit · session discipline) · Prompt Playbook build loop (P0–P26, conductor/builder pattern) · agent hooks & guardrails (secret-path write denial, post-commit security pass) · multi-agent runs (3 concurrent) + swarm experiments (Ruflo) · gstack skill suite · Obsidian LLM second-brain with automated capture pipeline · prompt engineering · MCP integration · YC Paxel build telemetry · Standout AI-native engineer ranking

Testing, QA & Security: Manual, functional & policy/compliance testing · UI automation (Maestro) · Competitive parity benchmarking · Test auditing · Property-based testing · JIRA · Agile · SLA-driven operations · OWASP Top 10 2025 & LLM Top 10 · gitleaks · bearer

Programming & Data: React Native (advanced) · TypeScript · Node.js · Python · PostgreSQL · Next.js · Android · React (intermediate) · Kotlin · iOS (beginner) · Java · C++ · SQL · HTML/CSS · Linux

Cloud & Platforms: Supabase (Postgres, RLS) · Railway · FastAPI + statsmodels · Expo + EAS · Capacitor · Jetpack Compose, Room & WorkManager · Android Health Connect · Deno edge functions · Node/Express · Redis · Paddle · Stripe · Razorpay · Apple IAP · WhatsApp Business API · Playwright / headless Chromium · WebSockets (live step streaming) · Sentry · PostHog · Docker · AWS (ECR, App Runner, RDS) · Amazon Bedrock · SQLAlchemy · Turborepo + pnpm · Prisma · pgvector · Git/GitHub

Growth & Marketing: Google Ads · Meta Ads · Programmatic SEO (pSEO) · Landing pages & funnels · Cold outreach (Instagram to Zoom) · Analytics

EDUCATION

MCA, Computer Science (2018 – 2020) · BCA (2015 – 2017) — Devi Ahilya Vishwavidyalaya (DAVV), Indore

2015 – 2020

AWARDS & COMMUNITY

Amazon Spot Awards & Jedi Ranger Awards (multiple cycles) · Campus Ambassador, IIM Bangalore "Vista" (2021) · Volunteer, Robin Hood Army (2016 – Present)